

Postune For LLM

Yeebo Xie

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1 Set Up

Make some setup first

State S: prompt + generated response so far

Action A: generate next token

Rewards R: Deterministic Like the answer to a math question is right or not

Transition probabilities: $\mathbb{T}(s' | s, a)$

Policy π_θ : Language Model

Rollout: $s_1- > a_1- > s_2- > a_2- > \dots- > s_{done}- > R$

2 Policy Gradient

The expectation of reward is the integral over the s on reward given s, and policy action.

$$\mathbb{E}[R] = \int \mathbb{P}(s) \pi_\theta(a|s) \mathbb{R}(s, a) d(s)$$

Take gradient of it:

$$\nabla_\theta \mathbb{E}[R] = \int \mathbb{P}(s) \nabla_\theta \pi_\theta(a|s) \mathbb{R}(s, a) d(s)$$

$$\nabla_\theta \mathbb{E}[R] = \int \mathbb{P}(s) \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) \mathbb{R}(s, a) d(s)$$

$$\nabla_\theta \mathbb{E}[R] = \mathbb{E}[\nabla_\theta \log \pi_\theta(a|s) \mathbb{R}(s, a)]$$

Sample prompt s, sample response from $\pi_\theta(a|s)$

Update parameters based on $\log \pi_\theta(a|s) \mathbb{R}(s, a)$

3 DPO

Start from the RLHF objective, max reward and penalty to a reference π_{ref}

$$J(\pi) = \max_{\pi} \mathbb{E}_{x, y \sim \pi(y|x)} [r(x, y)] - \beta D_{KL}(\pi(\cdot|x) || \pi_{ref}(\cdot|x))$$

Our goal is to find the policy π_r that maximizes this $J(\pi)$

3.1 Step 1: Expand the KL divergence

$$D_{KL}(\pi(\cdot|x)||\pi_{ref}(\cdot|x)) = \sum_y \pi(y|x) [\log \pi(y|x) - \log \pi_{ref}(y|x)]$$

Substitute that back into J_π :

$$J(\pi) = \sum_y \pi(y|x) [r(x, y)] - \beta \sum_y \pi(y|x) [\log \pi(y|x) - \log \pi_{ref}(y|x)]$$

simplify:

$$J(\pi) = \sum_y \pi(y|x) [r(x, y) - \beta \log \frac{\pi(y|x)}{\pi_{ref}(y|x)}]$$

We must also keep in mind that probabilities sum to 1:

$$\sum_y \pi(y|x) = 1$$

3.2 Step 2: Optimize over all possible π

We now want to find the probability distribution $\pi(y|x)$ that maximizes $J(\pi)$. That's a constrained optimization problem with the normalization constraint. We use a Lagrange multiplier $\lambda(x)$ for the constraint and form:

$$\mathcal{L}(\pi, \lambda) = \sum_y \pi(y|x) [r(x, y) - \beta \log \frac{\pi(y|x)}{\pi_{ref}(y|x)}] + \lambda(x) [1 - \sum_y \pi(y|x)]$$

3.3 Step 3: Take derivative and set to zero

Take derivative of $\mathcal{L}$ w.r.t. $\pi(y|x)$ and set to 0:

$$\frac{\partial \mathcal{L}}{\partial \pi(y|x)} = r(x, y) - \beta [\log \frac{\pi(y|x)}{\pi_{ref}(y|x)} + 1] - \lambda(x) = 0$$

solve for $\log \frac{\pi(y|x)}{\pi_{ref}(y|x)}$:

$$\log \frac{\pi(y|x)}{\pi_{ref}(y|x)} = \frac{1}{\beta} [r(x, y) - \lambda(x)] - 1$$

Exponentiate both sides:

$$\pi(y|x) = \pi_{ref}(y|x) \exp(\frac{1}{\beta} [r(x, y) - \lambda(x)] - 1)$$

The constants 1 and $\lambda(x)$ can be absorbed into a single normalization constant $Z(x)$, because they do not depend on y

$$\pi(y|x) = \frac{1}{Z(x)} \pi_{ref}(y|x) \exp\left(\frac{1}{\beta} [r(x, y)]\right)$$

where $Z(x)$ ensures the probabilities sum to 1 (for Lagrange).

$$\pi(y|x) \propto \pi_{ref}(y|x) \exp\left(\frac{1}{\beta} [r(x, y)]\right)$$

Intuition:

High-reward actions get exponentially higher probability.

But it's still shaped by the reference model's probabilities.

β acts like a temperature: small $\beta \rightarrow$ more concentrated on high rewards.

3.4 Invert the relation to express reward in terms of the policy

From that equation, take log and rearrange:

$$r(x, y) = \beta \log \frac{\pi(y|x)}{\pi_{ref}(y|x)} + \beta \log Z(x)$$

where $Z(x)$ is a normalization constant to make probabilities sum to 1.

3.5 Bring in preference data

Human preference data says: on the same prompt x , completion y_w (winner) is preferred over y_l (loser).

$$P(y_w > y_l|x) = \sigma(r(y_w, x) - r(y_l, x))$$

Now plug in our expression for $r(x, y)$ from Step 3:

$$r(y_w, x) - r(y_l, x) = \beta \left[\log \frac{\pi(y_w|x)}{\pi_{ref}(y_w|x)} - \log \frac{\pi(y_l|x)}{\pi_{ref}(y_l|x)} \right]$$

and you want to maximize the p:

$$\mathcal{L}_{DPO} = -\log \sigma\left(\beta \left[\log \frac{\pi(y_w|x)}{\pi_{ref}(y_w|x)} - \log \frac{\pi(y_l|x)}{\pi_{ref}(y_l|x)} \right]\right)$$